DISCRETE STRUCTURES

CMSC250

Chapter 15: Strong Induction

By Naren Ayinala

All preceding cases, recurrences, and constructive arguments



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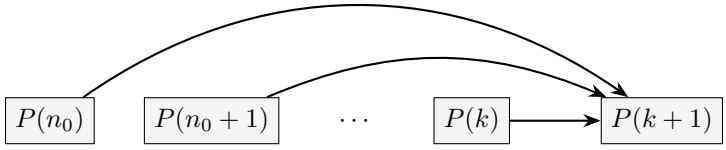
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1 Why ordinary induction sometimes feels insufficient

Ordinary induction assumes one preceding statement $P(k)$ to prove $P(k+1)$. Some recurrences or constructions depend on *several* earlier cases. **Strong induction** allows us to assume every statement from the base index through k.

Key idea. Strong induction: to prove $P(n)$ for every $n \geq n_0$, verify the initial case(s). Then for arbitrary $k \geq n_0$, assume $P(j)$ for *all* integers $n_0 \leq j \leq k$, and prove $P(k+1)$.



Strong induction is not logically stronger than ordinary induction; the two principles are equivalent. It is a more convenient way to organize certain proofs.

2 A complete strong-induction template

1. Define $P(n)$ precisely and identify its domain.
2. Verify the base case, or all needed initial cases.
3. Fix an arbitrary integer k in the required range. Assume $P(j)$ holds for every integer j from the starting index through k.
4. Prove $P(k+1)$, explicitly identifying which earlier indices are used and why they lie in the IH range.
5. Invoke strong induction and end with ■.

3 Prime factorization: the canonical example

Worked example: every integer at least two factors into primes

Question. Prove that every integer $n \geq 2$ is prime or can be expressed as a product of primes.

Proof. Let $P(n)$ be the stated property. **Base:** 2 is prime, so $P(2)$ holds. **Strong IH:** Fix an arbitrary integer $k \geq 2$. Assume that $P(j)$ holds for every integer j satisfying $2 \leq j \leq k$. **Step:** Consider $k+1$. If $k+1$ is prime, the claim is immediate. Otherwise, by the definition of composite, $k+1 = ab$ for integers a, b with $2 \leq a, b < k+1$. In particular, $2 \leq a, b \leq k$, so the strong IH applies separately to a and b. Each is prime or a product of primes; multiplying those representations expresses $k+1$ as a product of primes. Therefore $P(k+1)$ holds. By strong induction, the claim is true for every integer $n \geq 2$. ■

Why the proof works. We cannot predict that a composite $k+1$ factors using only the number k. We need the claim for the potentially much smaller factors a and b.

4 Recurrences that depend on two previous terms

A recurrence such as $a_n = a_{n-1} + a_{n-2}$ uses two earlier terms. To prove a statement about all its terms, it is natural to verify *two* base cases.

Worked example: Fibonacci terms grow at most exponentially

Question. Let $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. Prove $F_n \leq 2^n$ for all $n \geq 0$.

Proof. Let $P(n)$ assert $F_n \leq 2^n$. **Base cases:** $F_0 = 0 \leq 1$ and $F_1 = 1 \leq 2$. **Strong IH:** Fix arbitrary $k \geq 1$ and assume $F_j \leq 2^j$ for every integer $0 \leq j \leq k$. **Step:** Using the recurrence and the IH at $j = k$ and $j = k - 1$,

$$F_{k+1} = F_k + F_{k-1} \leq 2^k + 2^{k-1} \leq 2^k + 2^k = 2^{k+1}.$$

Both indices are in the IH range because $k \geq 1$. Thus $P(k+1)$ holds. Strong induction proves the bound. ■

5 Coin and postage constructions

A constructive strong-induction proof should say *which smaller construction* is reused and *which object* is added.

Worked example: forming all sufficiently large amounts using 3 and 5

Question. Prove that every integer $n \geq 8$ can be written $n = 3a + 5b$ for some nonnegative integers a, b.

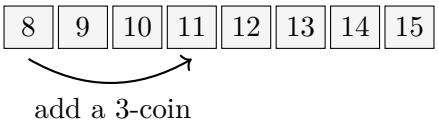
Proof. Let $P(n)$ assert that such a, b exist. **Base cases:**

$$8 = 3 + 5, \qquad 9 = 3 + 3 + 3, \qquad 10 = 5 + 5.$$

Thus $P(8)$, $P(9)$, and $P(10)$ hold. **Strong IH:** Fix arbitrary $k \geq 10$ and assume $P(j)$ holds for every integer $8 \leq j \leq k$. **Step:** Since $k + 1 - 3 = k - 2$ and $8 \leq k - 2 \leq k$, the IH gives nonnegative integers a, b such that $k - 2 = 3a + 5b$. Therefore

$$k + 1 = (k - 2) + 3 = 3(a + 1) + 5b.$$

The new coefficients $a + 1, b$ are nonnegative integers. Thus $P(k+1)$ holds, and strong induction completes the proof. ■



6 Why multiple base cases matter

Suppose a recurrence uses a_{n-2} and the induction step tries to prove $P(k+1)$ using $P(k-1)$. If the induction starts at $n = 1$, then the step $k = 1$ would require $P(0)$, which is outside the domain. Either prove enough initial cases and begin the step later, or change the domain correctly.

Worked example: a two-step inequality

Question. Let $a_0 = 1$, $a_1 = 2$, and $a_n = 2a_{n-1} + a_{n-2}$ for $n \geq 2$. Prove $a_n \leq 3^n$ for every $n \geq 0$.

Proof. Bases: $a_0 = 1 \leq 1$ and $a_1 = 2 \leq 3$. **Strong IH:** Fix arbitrary $k \geq 1$ and assume $a_j \leq 3^j$ for all $0 \leq j \leq k$. **Step:**

$$a_{k+1} = 2a_k + a_{k-1} \leq 2 \cdot 3^k + 3^{k-1} = 7 \cdot 3^{k-1} \leq 9 \cdot 3^{k-1} = 3^{k+1}.$$

Thus $P(k+1)$ holds. Strong induction gives the result. ■

7 An alternative: converting strong to ordinary induction

Let $Q(n)$ be the statement "$P(j)$ is true for every integer $n_0 \leq j \leq n$." If the strong-induction step proves $P(n+1)$ from all preceding cases, then an ordinary induction proof of $Q(n)$ works. This explains why strong induction does not grant extra logical power.

8 Common mistakes and repairs

- **Only one base for a two-step recurrence:** check enough consecutive starting indices.
- **Applying IH outside its range:** explicitly verify $n_0 \leq j \leq k$ before invoking $P(j)$.
- **Confusing existence with a formula:** for a coin problem, construct actual nonnegative coefficients.
- **Proving only examples:** a table suggests a pattern; the induction step establishes all remaining cases.
- **Assuming $P(k+1)$:** the strong IH stops at k.

9 Practice questions with complete solutions

Question. 1. Prove every integer $n \geq 12$ can be written as $4a + 5b$ for nonnegative integers a, b.

Solution. Base cases: $12 = 4+4+4$, $13 = 4+4+5$, $14 = 4+5+5$, $15 = 5+5+5$. Fix arbitrary $k \geq 15$ and assume the claim for every integer $12 \leq j \leq k$. Since $12 \leq k-3 \leq k$, write $k-3 = 4a+5b$ by the IH. Then $k+1 = 4(a+1)+5b$ with nonnegative integer coefficients. Strong induction completes the proof. ■**Question.** 2. Let $b_0 = b_1 = 1$ and $b_n = b_{n-1} + b_{n-2}$ for $n \geq 2$. Prove $b_n \leq 2^n$ for all $n \geq 0$.

Solution. Base cases: $b_0 = 1 \leq 1$ and $b_1 = 1 \leq 2$. Fix arbitrary $k \geq 1$ and assume $b_j \leq 2^j$ for all $0 \leq j \leq k$. Then $b_{k+1} = b_k + b_{k-1} \leq 2^k + 2^{k-1} \leq 2^{k+1}$. By strong induction, the result follows. ■**Question.** 3. Explain why proving only $P(8)$ is insufficient for the 3-and-5 coin proof above.

Solution. The step obtains $P(k+1)$ from $P(k-2)$. Starting at $k = 8$ would require $P(6)$, which is outside the claimed range. Three consecutive bases $8, 9, 10$ let the step begin at $k = 10$; then $k - 2 \geq 8$ and the IH applies.